Poster: Capture the List: Ranking Manipulation Leveraging Open Forwarders

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Abstract

Domain top lists are widely used in Internet measurement and security analysis among the Internet community. To improve the manipulation resistance and stability of top lists, many efforts have been devoted, such as the passive DNS (pDNS)-based top list, Secrank, and the aggregated list, Tranco. However, it remains unexamined how robust these newly-built top lists are against ranking manipulation. In this study, we study the pitfall of the pDNS data and propose **RaMOF**, a novel **Ra**nking **M**anipulation approach based on **O**pen **F**orwarders, leveraging the inherent forwarding relationships between open forwarders and public DNS services. Our experiments show that both pDNS-based top lists and aggregated lists are prone to be affected by RaMOF attacks.

CCS Concepts

• **Security and privacy** → **Network security**.

Keywords

Ranking manipulation, Top lists, Open forwarders, Forwarding relationship

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1 Introduction

Top lists are a collection of popular domains that serve as fundamental resources across a range of scenarios, including Internet measurement, security analysis, and threat detection. Traditional top lists have long been under scrutiny due to their susceptibility to ranking manipulation, passive DNS (pDNS)-based (e.g., Cisco Umbrella, Farsight and SecRank [4]) and aggregation of multiple top lists [1] are two promising directions of top lists with better manipulation resistance. pDNS-based top lists use DNS query traffic as input data to calculate the popularity of domains. Aggregated top lists generally combine several existing top lists and vote for ultimate domain ranking.

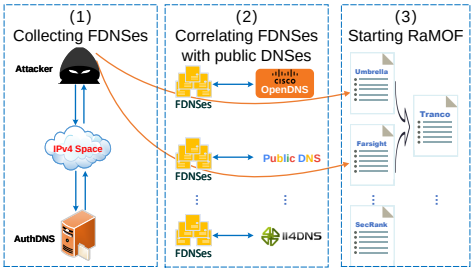

Figure 1: RaMOF Attack Overview.

In this study, we present RaMOF, a cost-effective method for ranking manipulation against newly-built pDNS-based top lists and aggregated top lists, including Secrank, Umbrella, Farsight, and Tranco. Our preliminary experiment results display that RaMOF can easily achieve a ranking of 4,899, 5,483, 500 ∼ 1,000 (estimated), and 1,360 in terms of pay-level domain in Umbrella, Secrank, Farsight, and Tranco, respectively.

2 RaMOF Attack

Pitfalls in Top Lists. 1) pDNS-based top lists rely on large-scale pDNS data from public DNS (e.g., Cisco OpenDNS), providing valuable insights into domain activity. We identified that the number of clients and domain query volume, two core metrics of pDNS-based top lists, are susceptible to manipulation. 2) Multiple homogeneous top lists would impair the manipulation resistance of aggregated top lists.

Attack Workflow. The basic idea of the RaMOF attack is to issue massive requests of a specific domain to a calculated set of open forwarders. The overview of the RaMOF attack is illustrated in Figure 1.

Step I. Collecting FDNSes. We use a similar method as illustrated in [5] that can observe FDNSes and indirect recursive resolvers (iRDNSes) on a single vantage point.

Step II. Correlating FDNSes with public DNSes. We first collect the backend resolver of the target public DNS [3]. Then, using the data observed in Step I, we enumerate the FDNSes that forward requests to a given public DNS.

Step III. Querying the target domain to collected FDNSes. These DNS traffic will be routed to the target public DNS, thus affecting the ranking of the target domain in the pDNS-based top lists.

The steps outlined above constitute the basic mode of the RaMOF attack, that is, the targeted mode (RaMOF-T). RaMOF-T implements

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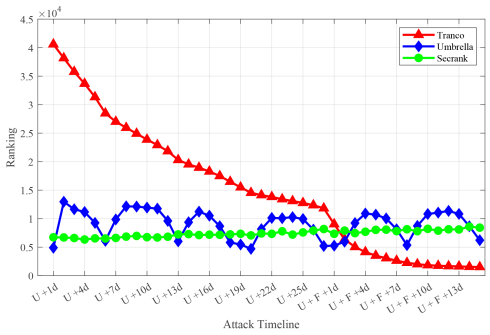

Figure 2: Ranking Trend across Multiple Top Lists.

a precise ranking manipulation attack by leveraging the forwarding relationship between open forwarders and the public DNSes. However, a lazy attacker probably skips **Step II** and issues queries to all the FDNSes he can discover, which we name as RaMOF attack in blind mode (RaMOF-B). Note that even when a top list undiscloses its data-collecting point, just as Farsight does, RaMOF-B still works.

3 Evaluation

Open Forwarder Population. According to [4], several factors would influence the ranking of specific domain, including the number of different request IPs, the quantity of the requests and the ranking algorithms. RaMOF tried to discover as many FDNSes as possible. We enumerated 410 and 493 iRDNSes of 114DNS and OpenDNS and correlated 251K and 531K FDNSes that forward queries to them, respectively. This means the RaMOF attack can trigger hundreds of thousands of FDNSes to generate artificial DNS queries to the data-collecting point of Secrank and Umbrella.

Evaluation of Ranking Manipulation. We managed to manipulate the rankings of our newly-registered test domain on 4 top lists using RaMOF attack, including Umbrella, Secrank, Farsight, and Tranco, using 20Mbps upstream bandwidth, which is common in residential networks.

Figure 2 shows the ranking trend of our domain in Secrank, Umbrella, and Tranco. When we launched the RaMOF-T attack against Umbrella and Secrank (*U+1d* ∼ *U+27d*), our domain reached a highest ranking of 4,899 in Umbrella and 5,483 in Secrank. During this stage, only one source top list of Tranco (i.e., Umbrella) was impacted, our domain achieved a rank of 11K in Tranco within 1 month's consecutive attack, proving that Tranco is better manipulation resistant than Umbrella. As we switched to the RaMOF-B attack (*U+F+1d* ∼ *U+F+15d*), the ranking of our domain in Tranco rocketed to 1,360 within 2 weeks, while the rankings remained relatively stable in Umbrella and Secrank compared to the RaMOF-T. The periodicity of rankings on Umbrella was aligned to previous research [2].

As Farsight's top list is not publicly available, we estimate the ranking of our domain in Farsight falls into 500 ∼ 1000 by reverse-engineering the ranking algorithm used by Tranco and the daily ranking in Umbrella and Tranco.

4 Conclusion

We uncover a systematic pitfall in modern pDNS-based top lists and homogeneous source lists vulnerability in aggregated top lists. We present RaMOF, an innovative ranking manipulation approach that leverages open forwarders, exploiting the inherent forwarding relationship between open forwarders and public DNS services. Our experiment results display that RaMOF can easily achieve a ranking of 4,899, 5,483, 500 ∼ 1, 000 (estimated), and 1,360 in terms of pay-level domain in Umbrella, Secrank, Farsight, and Tranco.

Acknowledgments

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